

AWS Redshift Project Documentation

Project Title:

Database and Table Creation in Amazon Redshift

Project Overview:

This project demonstrates the creation and management of a data warehouse using **Amazon Redshift**.

The objective was to create a database, define a table schema, insert data, and query the data using SQL.

Objectives

- Create a new database in Redshift
- Design and create a table
- Insert structured data into the table
- Retrieve and validate stored data
- Understand basic SQL operations in a cloud data warehouse

Services & Tools Used

Amazon Redshift (Serverless / Cluster)

Redshift Query Editor v2

SQL (Structured Query Language)

Architecture

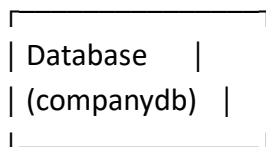
User (Query Editor v2)

|

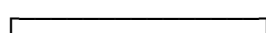


Amazon Redshift

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```
| Table      |  
| (employees) |  
└───────────┘
```

Implementation Steps

1 Connected to Redshift

- Accessed AWS Console
- Opened Redshift Query Editor v2
- Connected using temporary credentials

2 Created Database

```
CREATE DATABASE companydb;
```

3 Switched Database

- Changed connection from dev → companydb

4 Created Table

```
CREATE TABLE employees (  
    emp_id INT,  
    emp_name VARCHAR(100),  
    department VARCHAR(50),  
    salary INT  
);
```

5 Inserted Data

```
INSERT INTO employees VALUES  
(1,'Ravi','IT',50000),  
(2,'Arjun','HR',45000),  
(3,'Meena','Finance',60000);
```

6 Queried Data

```
SELECT * FROM employees;
```

Output:

Successfully created companydb database

Created employees table with defined schema

Inserted and retrieved records successfully

Verified data using SELECT query

Key Learnings:

Basics of Amazon Redshift architecture

Database and table creation in a data warehouse

SQL query execution in cloud environment

Handling errors (syntax issues, duplicate table creation)

Data validation and querying

Challenges Faced:

Syntax error due to trailing comma in table creation

Duplicate table creation error (relation already exists)

Duplicate data insertion

Conclusion:

This project provided hands-on experience in setting up and managing a cloud-based data warehouse using Amazon Redshift. It strengthened foundational knowledge of SQL operations and cloud database management.

Output image:

The screenshot displays the AWS Redshift Query Editor v2 interface. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/sqlworkbench/home?region=us-east-1#/client`. The interface includes a sidebar with navigation options: Editor, Queries, Notebooks, Charts, History, and Scheduled queries. The main editor area shows a SQL query: `SELECT * FROM employees`. The query is executed, and the results are displayed in a table format. The table has four columns: `emp_id`, `emp_name`, `department`, and `salary`. The results show six rows of data, with the first three rows being unique and the last three rows being duplicates of the first three. The status bar at the bottom indicates the query ID is 612380, the elapsed time is 274 ms, and the total rows are 6.

Query ID: 612380 Elapsed time: 274 ms Total rows: 6

emp_id	emp_name	department	salary
1	Ravi	IT	50000
2	Arjun	HR	45000
3	Meena	Finance	60000
1	Ravi	IT	50000
2	Arjun	HR	45000
3	Meena	Finance	60000